

2025 Division C Tournament - Univ. Of Georgia

Chemistry Lab Written Exam



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School Name/Team: _____

Student Names: _____

Instructions:

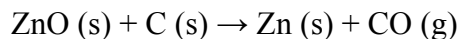
1. This test has a total of 35 points, displayed as a number in *italics* after a question.
2. Ties marked with a * in the bracket after the question, will be broken in order.

Part 1: Chemical Equations & Formula Writing

1. Write the chemical formulas for the following ions: bicarbonate, phosphate, and acetate. (3 points)

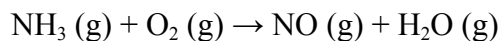
Part 2: Areas of Chemical Reactions/Stoichiometry

1. What type of reaction would this be classified as? (1 point)



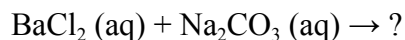
- A. Metathesis
- B. Acid-Base
- C. Combustion
- D. Redox

2. What are the coefficients for the balanced equation in order from left to right? (1 point)



- A. 2316
- B. 4546
- C. 1542
- D. 3614

3. Which of the following is true about this reaction in solution? Write the net ionic equation for this reaction and identify the spectator ions. (2 points)

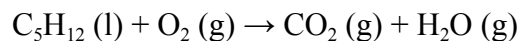


- A. $\text{Ba}^{2+} \text{ (aq)} + \text{CO}_3 \text{ (aq)} \rightarrow \text{BaCO}_3 \text{ (s)}$ is the net ionic equation.
- B. NaCl is aqueous.
- C. Na^+ and Cl^- are spectator ions.
- D. All of the above.

4. What is the molarity of a solution containing 3.794 g of AgNO_3 in 482 mL of water? (1 point)

- A. 0.0223 M
- B. 0.1056 M
- C. 0.0463 M
- D. 0.8031 M

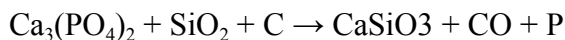
5. How many grams of carbon dioxide is produced from 65.28 g of pentane? (2 points)



- A. 199.0 g
- B. 139.81 g
- C. 63.69 g
- D. 97.80 g

6. You mix 50.0 mL of 0.100 M NaOH with 50.0 mL of 0.100 M HCl. What is the pH of the resulting solution after mixing? (3 points) [*]

7. Which of the following has the correct coefficients for the balanced equation? (2 points)



- A. 693827
- B. 491376
- C. 135352
- D. 289471

8. When a carbonyl is reacted with sodium borohydride, the carbonyl is reduced to an alcohol by the hydrogens of the borohydride. If the formula of sodium borohydride is NaBH_4 , how many grams of sodium borohydride are required to reduce 1.83 grams of 2-pentanone? (1 point)

- A. 1.61 g

B. 0.201 g

C. 0.804 g

D. 0.843 g

9. Below is a listed multistep reaction scheme:

R $\rightarrow$ 2B 86.3% yield

3B $\rightarrow$ 5G + C 53.0% yield

5C $\rightarrow$ 3F + 2D 71.9% yield

D $\rightarrow$ 4P 94.5% yield

If you begin with 10.0 moles of R, what is the total number of moles of G, F, and P combined?

(Don't round until the end) (2 points)

A. 19.3

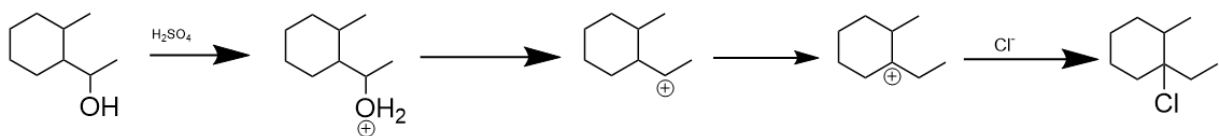
B. 18.2

C. 20.1

D. 19.9

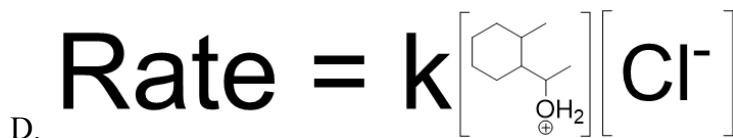
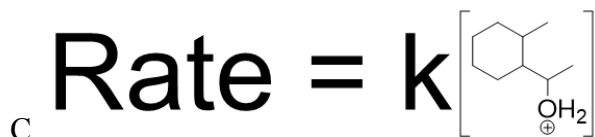
Part 3: Kinetics

1. Below is a simplified mechanism for a substitution reaction of an alcohol into an alkyl chloride. What is rate law based on the mechanism? (1 point)



A.
$$\text{Rate} = k \left[\text{1-methylcyclohexanol} \right]$$

B.
$$\text{Rate} = k \left[\text{1-methylcyclohexanol} \right] \left[\text{Cl}^- \right]$$

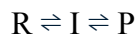


2. Determine the rate law for the following reaction (2 points) [*]:

[A] (M)	[B] (M)	Initial Rate (M/s)
0.10	0.10	2.0×10^{-4}
0.20	0.10	4.0×10^{-4}
0.10	0.20	8.0×10^{-4}

- A. $\text{Rate} = 2[\text{A}][\text{B}]^2$
- B. $\text{Rate} = 0.2[\text{A}][\text{B}]^2$
- C. $\text{Rate} = 0.2[\text{A}]^2[\text{B}]$
- D. $\text{Rate} = 2[\text{A}]^2[\text{B}]$

3. Assuming steady state conditions, the concentration of which molecule in the reaction below is considered constant? (1 point)



- A. R
- B. I
- C. P
- D. R and P

4. A change in which of the following factors does not change the value of the rate constant (k)? (1 point)

- A. Temperature
- B. Catalyst

- C. Reactant concentration
- D. Activation energy

5. What are the units of the rate constant (k) for a second-order reaction? (1 point)

- A. s^{-1}
- B. $M s^{-1}$
- C. $M^{-1}s^{-1}$
- D. $M^{-2}s^{-1}$

6. Which of the following statements is true? (2 points)

- A. The half life of a zeroth-order reaction is independent of concentration
- B. The half life of a second-order reaction is directly proportional to the square of the concentration
- C. The half life of a second-order reaction is directly proportional to the concentration
- D. The half life of a first-order reaction is independent of concentration

7. The rate determining step of a reaction is (1 point):

- A. The fastest step
- B. The first step
- C. The step with the smallest activation energy
- D. The step with the largest activation energy

8. What is meant by a pseudo-first order reaction? (2 points)

- A. A higher-order reaction where the concentration of one or more species in the rate law is high enough to be ignored.
- B. A higher-order reaction which depends on multiple molecules of the same species. ie: $\text{rate} = k[R]^2$
- C. A variable-order reaction where the order changes from second order to first order over time.
- D. A zeroth-order reaction that looks like a first order reaction when concentration is

graphed over time.

9. A reaction has rate constants $k = 2.50 \times 10^{-3} \text{ s}^{-1}$ at 300K and $k = 7.50 \times 10^{-3} \text{ s}^{-1}$ at 310K.

What is the activation energy? (2 points)

- A. 32.9 kJ/mol
- B. 45.6 kJ/mol
- C. 61.2 kJ/mol
- D. 91.2 kJ/mol

10. An enzyme speeds up a reaction by... (SELECT ALL THAT APPLY) (2 points)

- A. making the product more favorable, therefore forming it faster
- B. bringing substrate molecules closer together
- C. stabilizing the transition state
- D. adding more energy to the system

11. A reaction at equilibrium is given by: $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons \text{NH}_3(\text{g})$

In a sealed container at equilibrium, the partial pressures are: $P(\text{N}_2) = 0.50 \text{ atm}$, $P(\text{H}_2) = 0.30 \text{ atm}$, $P(\text{NH}_3) = 0.20 \text{ atm}$. Solve for K_p for this reaction. (2 points) [*]

- A. 3.04
- B. 2.96
- C. 2.60
- D. 1.82